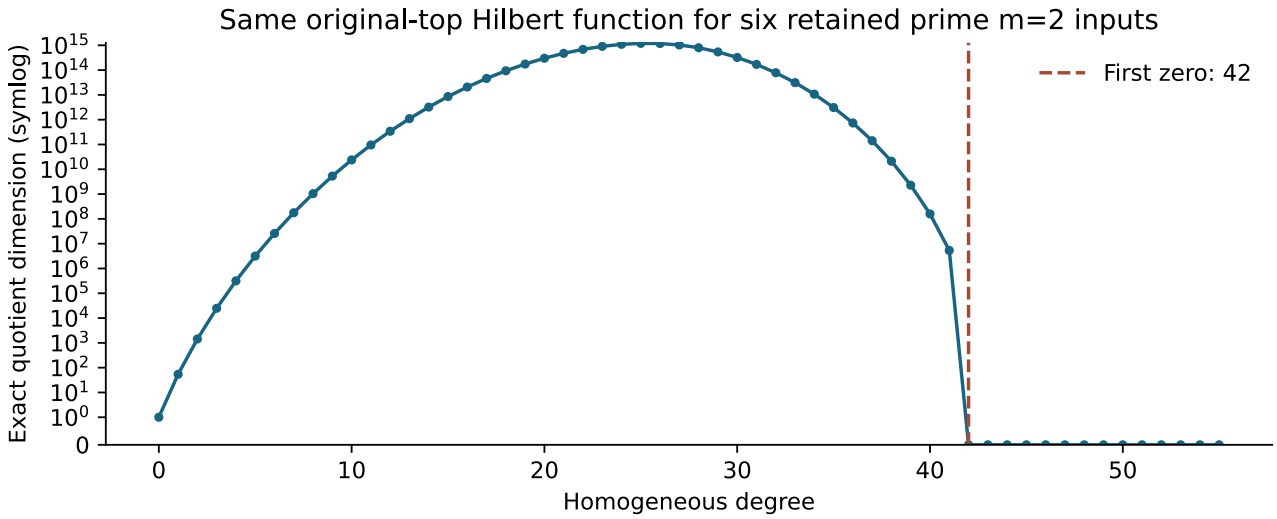
Prime m=2: exact original-top regularity

Observations only · n83-prime-m2-hilbert-20261028-r2

Six retained input identities, six producer certificates and six fresh-process replays passed. R2 had no failure, timeout, retry, resume or exclusion. R1 failed on its first source-template check; five cases were not reached. R1 data and partial results remain in r1-failed.tar.xz and experiment-hilbert-r1-failed.export.json.

Dataset	Targets	Exact degree
Original fixed base	Planted, natural	42
Paired baseline	Planted, natural	42
Paired spike40	Planted, natural	42

The quotient by the original highest homogeneous forms first vanishes in degree **42**. Its degree-41 coefficient is **5348880**; its degree-42 coefficient is zero. Total quotient dimension: **9159376218609600**. This is not a solving degree, last fall degree or affine final-basis degree.



Recorded exact Hilbert coefficients; symlog includes zero. All six source certificates identify this same function.

What was checked

The complete saved source matches the m2 summation polynomial and 54 selector equations. Invertible coordinate weights reduce the highest-form quotient to 54 variable squares and the product of two squared 27-variable sums. Wilson inclusion-matrix diagonals determine each multiplication rank; tensor images determine every bidegree coefficient. All 784 bidegrees are retained.

The prime p=48357032787619819080908 81 has a verified full Lucas certificate. Its characteristic exceeds 27. Controls passed: 168 explicit rank checks, 63 explicit product-degree checks and 44 small-characteristic rank deficiencies. Eight development unit tests passed.

Scope, proof and reproducibility

Proof scope

Lean 4.34.1 kernel-checked the finite first-zero arithmetic with no axioms. The subset-inclusion rank theorem, invertible substitution and the polynomial quotient/tensor-image bridge remain written mathematics, not Lean proofs. See HILBERT_DESIGN.md and HILBERT_PROOF_MAP.json. Fresh-process replay uses the producer implementation; independent scientific review is pending.

The six rows are saved identities, not independent random samples. Natural source files can coincide. Original and paired planted targets differ; paired planted tuples come from the common low26 support. The same fixed factor-base manifest is retained within each dataset across m2 through9.

Run identity

UTC start: 2026-10-28T16:40:30Z
UTC finish: 2026-10-28T16:40:33Z
Backend: existing-host fallback after retained EKS shared-bucket denial.
Limits: one thread, 2 GiB, 120 s each producer and replay; no timing claim.

Source: 64e06657f1a2e6df003130fa7f4e9a034d44b826
Protocol: 97da83f16a1a375a486546970b5b95c3313a63f838b366d07ff504209323a32b

Data and replay

The unchanged inputs, executable, every coefficient, rank diagonal, primality residue, control, log and replay receipt are preserved. The deterministic raw archive was downloaded from S3 into a fresh directory and all sealed file hashes checked. The source snapshot is separately retained.

Raw prefix: `s3://setec-astronomy-artifacts/runs/n83-prime-m2-hilbert-20261028-r2/local/`
Local: /home/ec2-user/n83-candidate-recovery/hilbert-r2
Repository: research/n83_candidate_pdp_20261028/evidence/hilbert/
Export: experiment-hilbert.export.json · Layout: HILBERT_FORMAT_V2.json
Archive: hilbert-archives/runs.tar.xz · Storage: evidence/hilbert-storage.json

Remaining work

Exact binary/Koblitz regularity and larger-m LFD remain open. Prime larger-m top quotients are positive-dimensional in the retained presentation; no finite vanishing degree is assigned. This batch identifies no elite base and makes no ECDLP attack, timing or universal FFD/LFD claim. All prior raw data remain.

Primary references

Ghorbani et al., Inclusion Matrices and Chains, Corollary 3: https://arxiv.org/pdf/0709.3144.
Phuong and Tran, A new proof of Stanley's theorem on the strong Lefschetz property, Theorem 2.7: https://arxiv.org/pdf/2211.13548.